

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Simulate cross correlation and autocorrelation on discrete time signals.	
Experiment No: 3	Date: 22-07-2026	Enrollment No: 92400133141

AIM: Simulate cross correlation and autocorrelation on discrete time signals.

Theory:

Cross-correlation and autocorrelation are mathematical operations used to measure the similarity or correlation between two signals. They are widely used in various applications, such as signal processing, image processing, and pattern recognition.

Cross-correlation measures the similarity between two signals at different time shifts. It computes the dot product of one signal with a time-shifted version of the other signal. The resulting cross-correlation signal indicates the similarity between the two signals at different time lags.

Autocorrelation, on the other hand, measures the similarity of a signal with a time-shifted version of itself. It computes the cross-correlation of a signal with itself. The autocorrelation signal shows how the signal is correlated with itself at different time lags.

Flowchart:

The flowchart for the program will consist of the following steps:
 Define two discrete-time signals (signal 1 and signal 2) or use the same signal for autocorrelation.

Compute the cross-correlation of the two signals or the autocorrelation of a single signal.

Plot and display the cross-correlation or autocorrelation signal.

Save the cross-correlation or autocorrelation signal (optional).

Now, let's see the Python program that simulates cross-correlation and autocorrelation on discrete-time signals:

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Program:

```

import numpy as np
import matplotlib.pyplot as plt
def cross_correlation(signal1, signal2):
# Compute the cross-correlation
    cross_corr = np.correlate(signal1, signal2, mode='full')
    return cross_corr
def autocorrelation(signal):
# Compute the autocorrelation
    auto_corr = np.correlate(signal, signal, mode='full')
    return auto_corr
# Define the discrete-time signals
signal1 = np.array([1, 2, 3, 4, 5])
signal2 = np.array([2, 4, 6, 8, 10]) # Compute the cross-correlation
cross_corr = cross_correlation(signal1, signal2) # Compute the autocorrelation
auto_corr = autocorrelation(signal1)
# Plot the cross-correlation and autocorrelation signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.stem(cross_corr)
plt.title('Cross-correlation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
plt.subplot(2, 1, 2)
plt.stem(auto_corr)
plt.title('Autocorrelation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
plt.tight_layout()
plt.show()
# Save the cross-correlation or autocorrelation signals (optional) # np.savetxt('cross_correlation.txt', cross_corr,
delimiter=',')
# np.savetxt('autocorrelation.txt', auto_corr, delimiter=',')

```

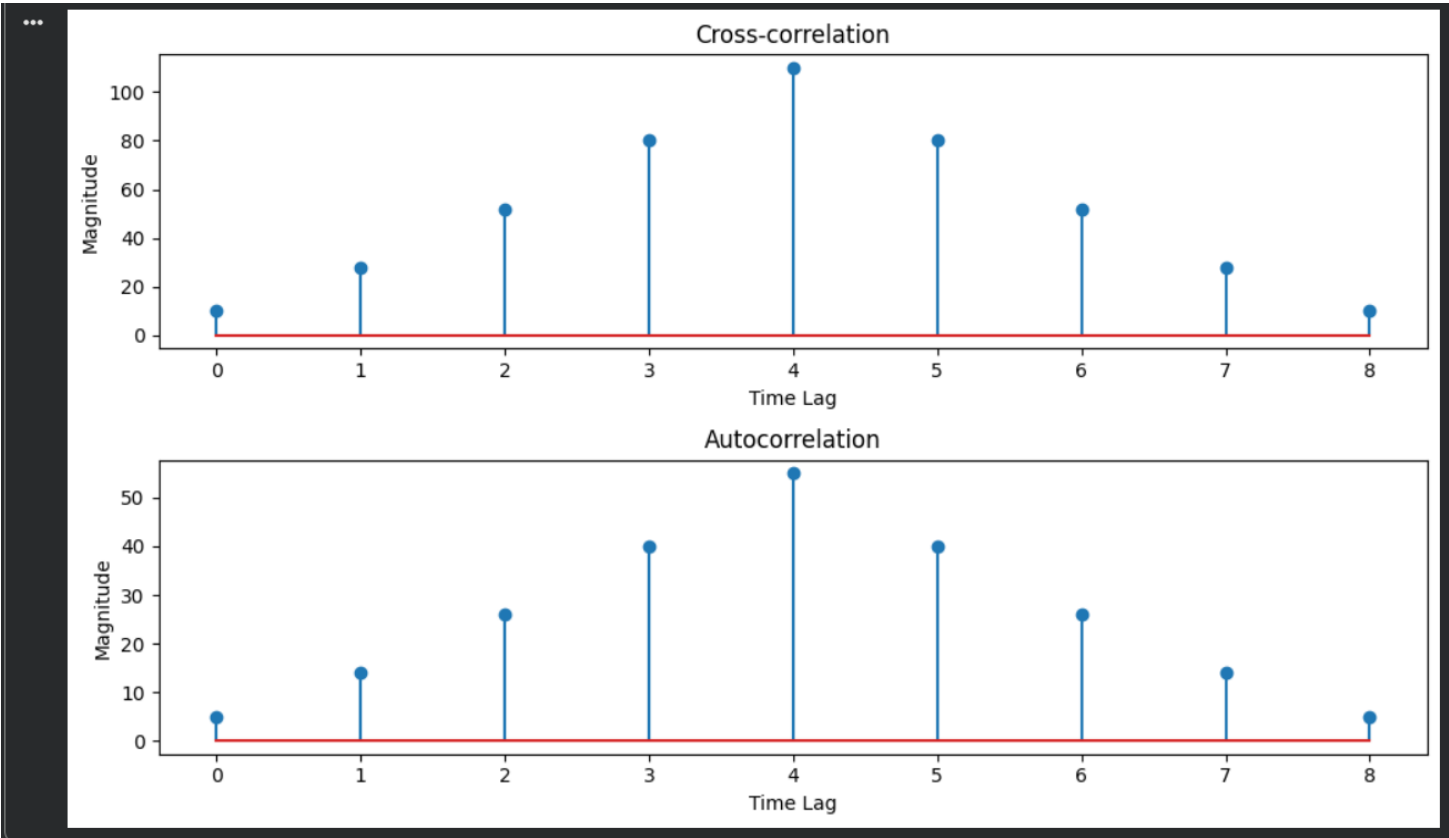
Expected Output:

A plot will be displayed showing the cross-correlation and autocorrelation signals.

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The cross-correlation and autocorrelation signals will be saved to text files (optional).

Feel free to experiment with different discrete-time signals and observe the cross-correlation and autocorrelation outputs.



What is Correlation?

- The correlation of two functions or signals or waveforms is defined as the measure of similarity between those signals. They are widely used in various applications, such as signal processing, image processing, and pattern recognition.

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- There are two types of correlations –
- Cross-correlation
- Autocorrelation

Cross-correlation

- The cross-correlation between two different signals or functions or waveforms is defined as the measure of similarity or coherence between one signal and the time-delayed version of another signal.

The cross-correlation between two different signals indicates the degree of relatedness between one signal and the time-delayed version of another signal.

- There are also many applications of signal crosscorrelation in signal processing systems, especially when the signal is corrupted by another undesirable signal (noise) so that the signal estimation (detection) from a noisy signal has to be performed.

Autocorrelation

- The autocorrelation function is defined as the measure of similarity or coherence between a signal and its time delayed version. Therefore, the autocorrelation is the correlation of a signal with itself.
- Physically, signal autocorrelation indicates how the signal energy (power) is distributed within the signal, and as such is used to measure the signal power. Typical applications of signal autocorrelation are in radar, sonar, satellite, and wireless communications systems. Devices that measure signal power using signal correlation are known as signal correlators.

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Given two discrete-time real signals (sequences) $x[k]$ and $y[k]$. The *autocorrelation and croosscorrelation* functions are respectively defined by

$$R_{xx}[k] = \sum_{m=-\infty}^{\infty} x[m]x[m-k], \quad R_{yy}[k] = \sum_{m=-\infty}^{\infty} y[m]y[m-k]$$

$$R_{xy}[k] = \sum_{m=-\infty}^{\infty} x[m]y[m-k], \quad R_{yx}[k] = \sum_{m=-\infty}^{\infty} y[m]x[m-k]$$

where the parameter k is any integer, $-\infty \leq k \leq \infty$.

Flowchart/Steps:

- The flowchart for the program will consist of the following steps:
- Define two discrete-time signals (signal 1 and signal 2) or use the same signal for autocorrelation.
- Compute the cross-correlation of the two signals or the autocorrelation of a single signal.
- Plot and display the cross-correlation or autocorrelation signal.
- Save the cross-correlation or autocorrelation signal (optional).

Code 2:

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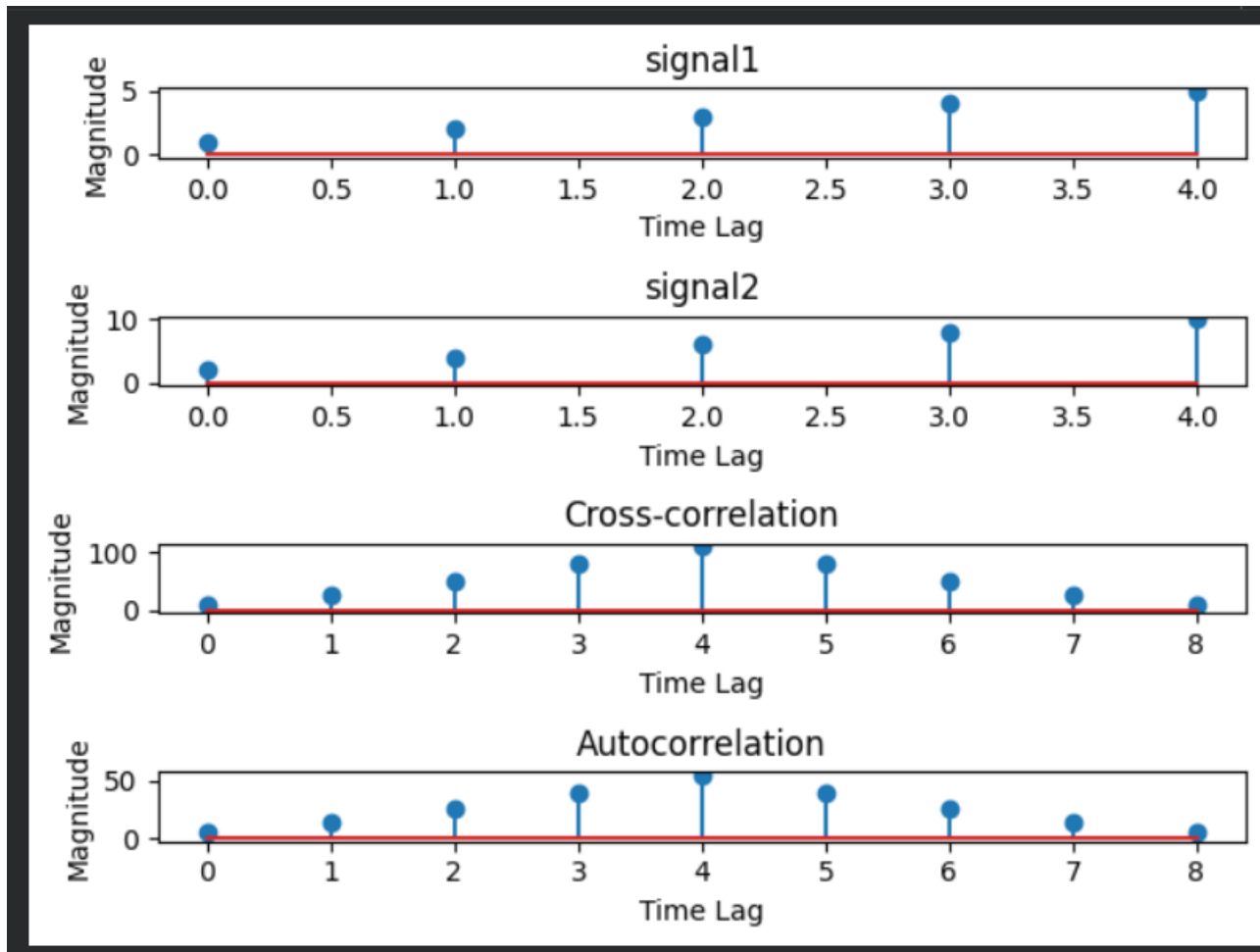
```

import numpy as np
import matplotlib.pyplot as plt
def cross_correlation(signal1, signal2):
# Compute the cross-correlation
    cross_corr = np.correlate(signal1, signal2, mode='full')
    return cross_corr
def autocorrelation(signal):
# Compute the autocorrelation
    auto_corr = np.correlate(signal, signal, mode='full')
    return auto_corr
# Define the discrete-time signals
signal1 = np.array([1, 2, 3, 4, 5])
signal2 = np.array([2, 4, 6, 8, 10]) # Compute the cross-correlation
cross_corr = cross_correlation(signal1, signal2) # Compute the autocorrelation
auto_corr = autocorrelation(signal1)
# Plot the cross-correlation and autocorrelation signals
# plt.figure(figsize=(10, 6))
# Signal 1
plt.subplot(4, 1, 1)
plt.stem(signal1)
plt.title('signal1')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude') # Signal 2
plt.subplot(4, 1, 2)
plt.stem(signal2)
plt.title('signal2')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude') # Cross-correlation
plt.subplot(4, 1, 3)
plt.stem(cross_corr)
plt.title('Cross-correlation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
# Autocorrelation
plt.subplot(4, 1, 4)
plt.stem(auto_corr)
plt.title('Autocorrelation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
plt.tight_layout()
plt.show()

```

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OUTPUT:



Objectives:

1. To understand the concept of correlation of discrete-time signals.
2. To simulate cross-correlation between two discrete-time signals using Python.
3. To simulate autocorrelation of a discrete-time signal using Python.
4. To study the similarity between signals at different time lags.
5. To plot and analyze cross-correlation and autocorrelation using stem plots.
6. To understand the practical applications of correlation in signal processing, pattern recognition, radar, sonar, and communication systems.

Conclusion:

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The experiment successfully simulated **cross-correlation and autocorrelation of discrete-time signals using Python**. Cross-correlation was used to determine the similarity between two different signals at various time shifts, while autocorrelation measured the similarity of a signal with its own shifted versions. The results show that the given signals have a strong correlation because Signal 2 is a scaled version of Signal 1. The autocorrelation reaches its maximum value at zero lag, demonstrating the highest similarity of the signal with itself. Thus, correlation is an important technique in **signal detection, pattern recognition, synchronization, radar, sonar, and communication systems**.